

EDUCATION

Memorial University of Newfoundland

Bachelor of Science (Honours) in Computer Science & Applied Mathematics

GPA: 3.76 / 4 — Computer Science GPA: 3.9 / 4 — Dean's List 2022–23

Selected coursework: Data Visualization; Data Structures & Algorithms; Web Development; Video Game Programming & AI Agents; Neural Networks; Calculus I, II & III; Linear Algebra; Numerical Analysis; Ordinary & Partial Differential Equations; Mathematical Modeling; Special & General Relativity

St. John's, NL

Sep 2021 – May 2025

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, Java, SQL

Databases: MongoDB

Frameworks: React, Node.js, Express.js, Next.js

Developer Tools: Linux, Git, Docker, Google Cloud Platform, VS Code, Neovim

Libraries & Technologies: Tailwind CSS, DaisyUI, Inngest, Pandas, NumPy, Matplotlib, scikit-learn, Three.js

EXPERIENCE

Computer Science Help Centre Assistant

Memorial University

Sep 2024 – Dec 2024

St. John's, NL

- Provide support to students with their computer science coursework and assignments
- Foster a collaborative and supportive learning environment

Teaching Assistant

Memorial University

Sep 2023 – Dec 2024

St. John's, NL

- Assist professors in grading assignments and exams for a 100-student mathematics class
- Deliver weekly tutorials to reinforce concepts and provide additional support to students
- Invigilate quizzes and exams, ensuring a fair and secure testing environment

Database Analyst

Memorial University

May 2022 – Aug 2022

St. John's, NL

- Developed a Selenium-based automation script to detect and resolve email delivery issues, improving communication reliability
- Saved 30 hours of staff time throughout the semester through process automation for the Co-operative Education Services Department

PROJECTS

Live Interview Platform | [Live Demo](#) | [GitHub](#)

Dec 2025

- Built a full-stack MERN application simulating real-world technical interviews with live coding and video communication
- Implemented LeetCode-style problem practice with an in-app code editor and automated code execution
- Enabled real-time interview sessions with secure authentication, video calling, and session management
- **Tools Used:** MongoDB, Express.js, React, Node.js, Tailwind CSS, DaisyUI, Stream, Clerk, Inngest

YouTube Clone | [GitHub](#) | Personal Project

Aug 2024

- Cloud-native architecture with Google Cloud Run for serverless deployment, Pub/Sub for message queuing, and Storage buckets for scalable media storage
- Backend logic powered by Firebase Functions
- Modern, type-safe front-end experience using the Next.js framework with TypeScript
- **Tools Used:** TypeScript, Next.js, Google Cloud Run, Pub/Sub, Storage buckets, Firebase Functions

Honours Thesis: Black Hole Stability (MOTS) | [Thesis](#)

Sep 2024 – Apr 2025

- Developed symbolic and numerical spectral methods to study MOTS stability in black hole spacetimes
- Validated eigenvalue computations against exact solutions, achieving relative errors as low as 10^{-5}
- Analyzed eigenvalue spectrum under parameter variation to identify critical stability transitions